

Pei Yang

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Education

National University of Singapore

PhD – Computer Science

- Supervisor: Prof. Mike Zheng Shou

Singapore

Aug 2023 – Current

(Expected Aug 2027 Graduation)

UESTC + University of Glasgow

BEng – Electronic Information Engineering (Honours, 4%)

BEng – EEE (1st Class Honours)

- GPA 3.96/4, ranking 2/242 (within top 1%)

Chengdu, China

Sep 2019 – Jun 2023

Publications

Computer Use Agents

macOSWorld: A Multilingual Interactive Benchmark for GUI Agents

[Pei Yang*](#), [Hai Ci*](#), Mike Zheng Shou

NeurIPS 2025 [Paper](#) | [Presentation](#) | [Project Page](#) | [Code](#) | [Product](#)

In-Context Defense in Computer Agents: An Empirical Study

[Pei Yang*](#), [Hai Ci*](#), Mike Zheng Shou

2025 [Paper](#)

Diffusion Safety & Security

RingID: Rethinking Tree-Ring Watermarking for Enhanced Multi-Key Identification

[Hai Ci*](#), [Pei Yang*](#), [Yiren Song*](#), Mike Zheng Shou

ECCV 2024 [Paper](#) | [Project Page](#) | [Code](#)

Can Simple Averaging Defeat Modern Watermarks?

[Pei Yang*](#), [Hai Ci*](#), [Yiren Song](#), Mike Zheng Shou

NeurIPS 2024 [Paper](#) | [Presentation](#) | [Code](#)

IDProtector: An Adversarial Noise Encoder to Protect Against ID-Preserving Image Generation

[Yiren Song*](#), [Pei Yang*](#), [Hai Ci](#), Mike Zheng Shou

CVPR 2025 [Paper](#) | [Code](#)

WmAdapter: Adding Watermark Control to Latent Diffusion Models

[Hai Ci](#), [Yiren Song](#), [Pei Yang](#), [Jinheng Xie](#), Mike Zheng Shou

ICML 2025 [Paper](#)

DiffSeg30k: A Multi-Turn Diffusion Editing Benchmark for Localized AIGC Detection

[Hai Ci*](#), [Ziheng Peng*](#), [Pei Yang](#), [Yingxin Xuan](#), Mike Zheng Shou

2025 [Paper](#)

Embodied AI

ActionMap: Robot Policy Learning via Voxel Action Heatmap

[Pei Yang*](#), [Hai Ci*](#), [Yanzhe Chen*](#), [Qi Lv](#), [Han Cai](#), Mike Zheng Shou

2026 (Under review) [Paper](#) | [Project Page](#)

X-Humanoid: Robotize Human Videos to Generate Humanoid Videos at Scale

Pei Yang^{*}, Hai Ci^{*}, Yiren Song, Mike Zheng Shou

2025 (Under review) [Paper](#) | [Project Page](#) | [Presentation](#) | [Post](#)

H2R-Grounder: A Paired-Data-Free Paradigm for Translating Human Interaction Videos into Physically Grounded Robot Videos

Hai Ci^{*}, Xiaokang Liu^{*}, Pei Yang, Yiren Song, Mike Zheng Shou

2025 (Under review) [Paper](#) | [Project Page](#)

UENR-600K: A Large-Scale Physically Grounded Dataset for Nighttime Video Deraining

Pei Yang^{*}, Hai Ci^{*}, Beibei Lin, Yiren Song, Mike Zheng Shou

2026 (Under review) [Paper](#) | [Project Page](#)

Experiences

NVIDIA Research Singapore

Singapore

Research Collaboration – Efficient AI

Apr 2026 – Jun 2026

- Supervisor: [Han Cai](#)
- Project 1: [ActionMap](#) (+8.2% on LIBERO, +43.3% on real-world Franka)
- Project 2: Agentic Image Generation (built data pipeline + functional agentic system prototype)

Talks / Presentations

Walking Through macOSWorld

Apr 2026

Tech Share at Tencent Singapore [Presentation](#)

Invited Talk: Digital Watermarking in the Diffusion Era

Sep 2024

School of Computing Media Lunch Talk [Slides](#)

Tutorial: Video Diffusion Models

Nov 2023

Mike Zheng Shou, Pei Yang, Jay Wu

[Video](#) | [Slides](#)

Honours & Scholarships

2025	NeurIPS 2025 Top Reviewer
2023	Sichuan Provincial Outstanding Graduate (四川省优秀本科毕业生)
2022, 2021, 2020	Academic (Half-Tuition) Scholarship, Glasgow College UESTC (6%) (×3)
2022, 2021, 2020	Outstanding Student Scholarship (1 st Class), UESTC (10%) (×3)
2022, 2020	James Watt Scholarship of Innovation, Glasgow College UESTC (6%) (×2)
2022	Suzhou Industrial Zone Scholarship, UESTC (1%)
2020	Wuliangye Scholarship, UESTC (0.25%)

Skills

Programming	Python, C/C++, MATLAB, AppleScript
Computing	PBS, Slurm, Singularity
Technical Skills	PyTorch, TensorFlow, FFmpeg, WSL, AWS EC2, Franka, Claude Code
Software	Unreal Engine, Blender, VMware, Adobe Ps/Pr/Ae/Au, DaVinci Resolve